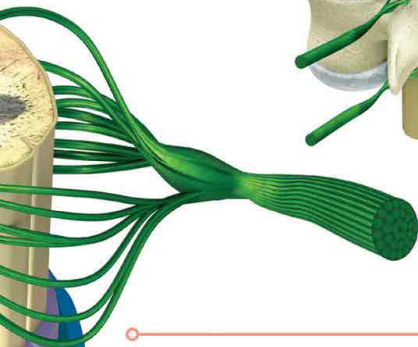
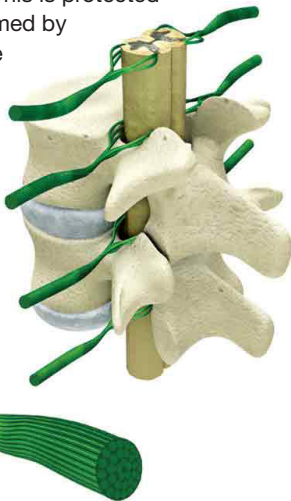


Protecting the spinal cord

The bundle of neurons that forms the spinal cord is made of soft tissue. This is protected by a tunnel of bone formed by the vertebrae that make up the backbone. These vertebrae are separated by thick disks of cartilage.



Information highway

A cross-section through the spinal cord shows butterfly-shaped gray matter in the center. The gray matter transmits signals across the cord. These signals are received from, and passed to, spinal nerves, which provide the link to the rest of the body. The white matter relays signals to and from the brain.

WITHOUT THINKING

Reflexes are automatic actions that happen without our being aware of them. Many reflexes, such as this withdrawal reflex, protect the body from danger and are controlled by the spinal cord. Reflex actions happen quickly because nerve signals travel through the spinal cord without going to the brain.



Danger

Pain receptors detect the burning heat of the flame and send signals to the spinal cord.



Withdrawal

The spinal cord sends signals to an arm muscle that pulls the hand away from the candle.



Pain

A message is now sent up the spinal cord to the brain and the person feels pain.